

NICK FAUSTI

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SUMMARY: A pragmatic and well-rounded software engineer with 10+ years of dynamic full stack software engineering experience. **MSE from University of Pennsylvania in Computer and Information Science.** Roles have encompassed positions in **software engineering, applied machine learning, product development, laboratory information management systems, natural language processing, and data science.** Expert command of a spectrum of programming technologies, development environments, and best practices in object-oriented development and testing. Proven track record of delivering on both planned and unplanned efficiencies. Able to work effectively in team-based environments utilizing the agile development methodology.

PROFESSIONAL EXPERIENCE:

Google – Mountain View, CA

Senior Software Engineer

Oct 2024 – Present

- Part of a team bringing Gemini to Google Assistant
- Co-invented (patent pending) LLM-based defect detection and management system for Google Home and Assistant, expanding to all Google products
- Worked on feature development and bug fixes for Gemini for iOS app

LANGLEY AI – heylangley.com

Founder & Software Engineer

Feb 2023 – Mar 2024

- Built an LLM-based language learning and speaking partner web application in Next.JS as a side project
- Implemented conversational memory and retrieval using Postgres vector stores and cloud functions
- Supported over 100 initial users, obtaining feedback and implementing feature requests quickly
- Open-sourced and made freely available at heylangley.com

GfK – Remote

Software Engineer

Mar 2023 – Oct 2024

- Led a team of engineers in developing and maintaining features for client-facing Insights market research data science platform using Node, React, Redux, TypeScript, and Azure
- Migrated system-wide state management/data-fetching solution from standard Redux to Redux + RTK Query to improve speed and efficiency of frontend and API calls
- Migrated multiple projects from webpack to Vite build systems
- Built an interactive, client-facing SVG/D3-based correspondence analysis and data visualization/charting tool
- Implemented NLP search using open-source LLM over our catalog of questionnaire datasets to make knowledge discovery and retrieval easier and interactive

CLYDE – Remote

Lead Software Engineer

Apr 2022 – Dec 2022

- Led a team of remote software engineers to develop and maintain client facing applications used by +100k people
- Ongoing full-stack development and engineering management
- Launched multiple features from idea to implementation and maintenance with React, Node, TypeScript and AWS

smileML – Remote

Senior Software Engineer

Jun 2021 – May 2022

- Developed two desktop application minimum viable products (MVPs) in one month using Electron, React, Tailwind, and Firebase
- Ongoing full-stack development and engineering management for cross-platform machine learning based applications
- Integrate with Stripe/PayPal to allow customers to sign up for additional features and services from within the app
- Developed a calendar synchronization system and data pipeline to integrate and launch application based on calendar events and attendants

INTELLECTUS APP – Los Angeles, CA

Technical Advisor

Mar 2020 – May 2021

- Architected and developed an MVP of an online learning platform in 3 weeks in vanilla JS, Flask, and Open-CV
- Computer vision-based web application to help keep students engaged in class and increase retention of material
- App is currently in use by NASA JPL and California public schools

BRIGHTFIELD GROUP LLC – Chicago, IL (Remote)

Sr. Software Engineer - React & Django

Jun 2017 – Mar 2020

- Developed a full-stack single-page application dashboard in React/Redux + Django Rest Framework to view analytics information related to natural language processing pipelines
- Integrated various social media APIs into SPA backend
- Worked with vast amounts of data; wrangling, cleaning, and deploying into analytics systems
- Built an NLP topic modeling/machine learning model and pipeline using sk-learn and pandas/pyspark
- Scraped content from the web, regularized, cleaned, and stored in backend to be used in offline machine learning processing
- Built an openCV landing page that takes in user's photos and returns color/mood related information for each image
- Built a marketing landing-page and social login data collection CMS

IQVIA – Plymouth Meeting, PA (Remote)

Associate Director - Software Engineering — Analytics Center of Excellence

Jun 2018 – Apr 2019

- Created a React/Redux front-end to display results and allow users to interactively apply changes to model parameters of ML API
- Built an API in Python/Django Rest Framework to expose machine learning models that help facilitate the selection of clinical trial investigator sites

UNIVERSITY OF PENNSYLVANIA – Perelman School of Medicine, Philadelphia, PA

Sr. Research Programmer – Positive Psychology Center - World Well-Being Project

Jun 2016 – Jun 2017

- Developed a React single-page application front-end for a personality evaluation tool in React
- Scraped all hospital reviews from Yelp using Scrapy library to run differential language analysis on
- Ran differential language analysis on multiple hospital characteristics using yelp review language data
- Building web infrastructure to expose internal natural language processing APIs to external researchers

Application Developer – Center of Excellence in Environmental Toxicology

Dec 2011 – May 2016

- Created a tool to automate the visual selection and data analysis of the calcium signaling fluorescence variation in cells as a result of pharmacological treatments using OpenCV, matplotlib, and Python
- Created a web based mapping and search system with Google Maps to help with the inference of a correlation between healthcare utilization in Northeastern Pennsylvania and hydraulic natural gas fracturing nearby
- Developed a freezer/reagent inventory application with Django, DRF API, and iOS barcode scanning front-end
- Worked as part of a team to create a natural language search system using the Python NLTK library to allow scientists to search medical records, environmental exposure data, and retrieve blood/tissue sample information for those records in plain English queries
- Successfully managed the migration of tissue sample data and participant case report form data into a new laboratory information management system at the proof of concept stage
- Gathered requirements for lab workflows and processes to be used as a model for future data migrations throughout the Perelman School of Medicine/Hospital of the University of Pennsylvania
- Working on creating an ontological framework to be used as a data standard for biomedical investigations throughout the Perelman School of Medicine
- Working on an iOS application for principal investigators to use as a tool to stay informed on the state of various projects/studies and view results from experiments involved with each study

SINGLE THROW INTERNET MARKETING, Wall, NJ

.NET Application Developer

Jun 2010 – Nov 2011

- Developed a content management system to offer to new and prospective clients. Allowed our clients with little technical knowledge to make simple content updates to their websites
- Trained clients and Single Throw employees to use the new content management system
- Build and manage web applications and projects using ASP.NET 1.1-3.5 framework
- Supervise a small team of developers that maintain and modify existing sites for over 200 clients
- Variety of skills: C#/VB.NET, ASP.NET Web Forms, Microsoft SQL Server 2005/2008, IIS6/7/, T-SQL, Stored Procedures, XML/XSLT/XPath, JavaScript, jQuery, HTML/CSS, Server/Database Migrations & Deployments

COMCAST CORP., Mt. Laurel, NJ

Software Test Engineer

Jan 2010 – Jun 2010

- Manual and automated testing of customer account management software
- Worked with development teams during all code deployments to ensure successful transitions from staging to production environments
- Gained exposure to all areas of the software development life cycle

JOHNSON & JOHNSON – Information Technology Services, Raritan, NJ

Software Development Intern

Jun 2008 - Sep 2009

- Created a web form application that greatly increased the productivity of Enterprise Data Backup team members
- Implemented changes and updates to current applications at the request of EDB team members
- Gained invaluable experience working in a team based environment as well as practical technical knowledge

PERSONAL: Eniapp Inc.

Nov 2013 - Present

Founder / Software Engineer

- Started a software consulting company that specializes in web development, NLP, and machine learning

- <https://eniapp.io>

Github Profile: <https://github.com/ncfausti>

EDUCATION: **Master of Science in Engineering - Computer and Information Science**

University of Pennsylvania, School of Engineering and Applied Science, Philadelphia, PA

BA in Information Technology & Informatics; Economics Minor

Rutgers University – School of Arts and Sciences, New Brunswick, NJ

CORE SKILLS: JavaScript/HTML5/CSS3; React; Python; Java; AWS; Google Cloud Services; Web Scraping; Data Science; Machine Learning; NLP, Unreal Engine, Unity, Computer Vision, Deep Learning

PUBLICATIONS:

Online Ratings of the Patient Experience: Emergency Departments Versus Urgent Care Centers Anish K Agarwal, Kevin Mahoney, Amy L Lanza, Elissa V Klinger, David A Asch, Nick Fausti, Christopher Tufts, Lyle Ungar, Raina M Merchant

Annals of Emergency Medicine, Volume 73, Issue 6, 631 - 638

Patient Experience and Satisfaction in Online Reviews of Obstetric Care: Observational Study Emily K Seltzer, Sharath Chandra Guntuku, Amy L Lanza, Christopher Tufts, Sindhu K Srinivas, Elissa V Klinger, David A Asch, Nick Fausti, Lyle H Ungar, Raina M Merchant

JMIR Formative Research 6 (3), e28379

Unconventional Gas and Oil Drilling Is Associated with Increased Hospital Utilization Rates Thomas Jemielita, George L Gerton, Matthew Neidell, Steven Chillrud, Beizhan Yan, Martin Stute, Marilyn Howarth, Pouné Saberi, Nicholas Fausti, Trevor M Penning, Jason Roy, Kathleen J Propert, Reynold A Panettieri Jr

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REFERENCES: Furnished upon request.